

- RAG Learning

Required (Core) Practicals — CP1 to CP8

Edexcel GCSE (9-1) Physics (1PH0) — Assessed in the written exams

Edexcel GCSE Physics specifies eight mandatory Core Practicals (CPs). The specification labels each one by its specification point (shown in brackets below); CP1–CP8 simply follows the specification's order. You will not be assessed on performing them in an exam, but you **must** know the method, variables, measurements, analysis, sources of error and improvements for each. They belong to Topics 2, 4, 5, 10, 14 and 15.

CP1 (2.19) — Investigating force, mass and acceleration using trolleys

Apparatus: Trolley, ramp (runway) with a wooden block to tilt it, bench pulley and string, slotted masses on a hanger, extra masses to add to the trolley, metre rule, masking tape or chalk, stopwatch (or two light gates and a data logger), balance.

Method

1. Tilt the ramp very slightly so the trolley just keeps rolling at a steady speed once pushed — this compensates for friction.
2. Mark a start line and two further lines a measured distance apart (e.g. 20 cm and 80 cm down the ramp).
3. Tie a string to the trolley, pass it over the pulley at the end of the ramp and hang a hanger with slotted masses on it. The weight of the hanging masses ($W = mg$) is the accelerating force.
4. Release the trolley from rest at the start line. Time how long it takes to pass each marked line (or let the light gates record the speed at each gate and the time between them).

5. Calculate the acceleration: $a = (v - u) / t$, using the average speed over each short section, or directly from the light-gate data.
6. Keep the accelerating force the same and add masses to the trolley one at a time; measure the total mass on a balance and repeat. (To study force instead, move masses from the trolley to the hanger so the total mass stays constant.)
7. Repeat each run three times and take a mean.

$$F = m \times a \quad a = (v - u) / t$$

Analysis

- For a constant force, acceleration decreases as the mass increases: a is inversely proportional to m , so a graph of a against $1/m$ is a straight line through the origin.
- For a constant total mass, a graph of acceleration against force is a straight line through the origin; its gradient is $1 / \text{mass}$.
- The mass being accelerated includes the hanging masses as well as the trolley.

⚠ Safety: Place a box or cushioned stop at the bottom so falling masses and a fast trolley are caught; keep feet clear of the hanging masses.

Exam Tip: Light gates are better than a hand-held stopwatch because they remove human reaction time. Always say what the independent, dependent and control variables are (e.g. keep the ramp angle and the force the same when mass is varied).

CP2 (4.17) — Measuring the speed, frequency and wavelength of waves in a fluid and a solid

Apparatus: Ripple tank with lamp and screen, motor-driven dipper connected to a low-voltage supply, stroboscope or phone camera, metre rule, stopwatch; metal rod, clamp stand with rubber-padded clamp, hammer, microphone connected to an oscilloscope or a frequency app.

Ripple tank method (waves in a fluid)

1. Set up the tank with water to a constant depth (about 1 cm). Turn on the dipper; observe the ripples on a screen beneath.
2. Measure **frequency**: count the number of waves passing a fixed point in 10 s and divide by 10 (or adjust the stroboscope until the pattern appears frozen — the strobe frequency then equals the wave frequency).
3. Measure **wavelength**: photograph the pattern next to a ruler (or use the frozen image), measure across ~5 wavelengths and divide by the number of waves.
4. Calculate wave speed $v = f \times \lambda$. You can also time one wave front over a measured distance: $v = \text{distance} / \text{time}$.

Metal rod method (waves in a solid)

1. Support the rod horizontally at its centre in the padded clamp so it can vibrate freely.
2. Hit one end of the rod firmly with the hammer; hold the microphone near the other end.
3. Read the peak (loudest) frequency from the oscilloscope or frequency app.
4. Measure the length of the rod. For the main vibration, the wavelength is twice the length of the rod ($\lambda = 2L$).
5. Calculate $v = f \times \lambda$ and repeat several times for a mean.

$$v = f \times \lambda$$

Suitability of equipment

- Water waves are slow and have low frequency, so they can be seen, counted and photographed directly.
- Sound waves in a solid travel several thousand m/s with frequencies of kHz, far too fast to count by eye — an electronic sensor (microphone and oscilloscope/app) is needed.

⚠ Safety: Mop up spilt water straight away and keep the ripple-tank power supply and lamp away from water. Keep fingers clear when striking the rod.

Exam Tip: Measure across several wavelengths to reduce uncertainty — the % error in a single wavelength reading is large.

CP3 (5.9) — Investigating refraction of light through rectangular glass blocks

Apparatus: Rectangular glass block, ray box with slit, protractor, white paper, sharp pencil, ruler.

Method

1. Place the block on white paper; draw round it.
2. Shine a ray into the block at an angle. Mark three points on the incident ray before the block and three after exit.
3. Remove the block; draw the incident, refracted and emerging rays.
4. Draw the normal at the point of entry; measure angle of incidence (i) and angle of refraction (r).
5. Repeat for at least 6 different values of i from 10° to 70° .
6. Plot $\sin i$ against $\sin r$ — the gradient = refractive index (extension).

What you should see

- Entering the glass (air $\rightarrow$ glass), light slows down and bends **towards** the normal, so r is smaller than i .
- Leaving the glass (glass $\rightarrow$ air), light speeds up and bends **away from** the normal. The emerging ray is parallel to the incident ray.
- A ray along the normal ($i = 0^\circ$) passes straight through without changing direction.

⚠ Safety: The ray box gets hot — let it cool before handling. Work in a dimmed room carefully, keeping bags and stools out of walkways.

Exam Tip: Always measure angles **from the normal**, not from the surface. Use a sharp pencil for accurate ray lines.

CP4 (5.19P) — How the nature of a surface affects thermal energy radiated or absorbed

Apparatus: At least four identical beakers (or boiling tubes) covered in different surfaces — e.g. matt black paper, shiny black, white paper, shiny aluminium foil — lids (card), kettle of hot water, measuring cylinder, thermometers, stopwatch.

Method (radiating)

1. Cover each container with a different surface material.
2. Use a measuring cylinder to pour the same volume of hot water (e.g. 200 cm³) into each container and put a lid on each.
3. Record the starting temperature of each, then record the temperature every minute for about 10 minutes using the stopwatch.
4. Calculate the temperature drop for each container, or plot temperature against time on the same axes.

Absorbing (variation)

- Fill the covered containers with the same volume of cold water and place them the same distance from a radiant heater or lamp. The container whose temperature rises fastest has the best absorbing surface.

Results

- **Matt black** surfaces are the best emitters and the best absorbers of infrared radiation — the water cools (or warms) fastest.
- **Light, shiny** surfaces (e.g. silver foil) are the poorest emitters and absorbers (they reflect infrared) — slowest temperature change.

Control Variables

- Volume and starting temperature of water;
- Size, shape and material of the container, and use of a lid;
- Time between readings and room temperature (keep away from draughts).

⚠ Safety: Take care pouring hot water; stand containers on a heatproof mat and do not carry them when full.

Exam Tip: Start all containers at the same temperature: the rate of cooling depends on the temperature difference with the surroundings, so a hotter start would give an unfair comparison.

CP5 (10.17) — Electric circuits: V , I and R for a resistor and a filament lamp; series and parallel circuits

Apparatus: Cell(s) and battery pack (or variable DC power supply), ammeters, voltmeters, fixed resistor, filament lamps, variable resistor (rheostat), connecting leads and switch.

Part a — resistor and filament lamp

1. Connect the component in series with the ammeter and variable resistor.
2. Connect the voltmeter in parallel across the component.
3. Adjust the variable resistor to get a range of currents; record V and I at each.
4. Reverse the connections to the cell to get negative values of V and I ; repeat readings.
5. Replace the resistor with a filament lamp and repeat.
6. Plot I (y-axis) against V (x-axis) and calculate $R = V / I$ for each pair of readings.

$$V = I \times R$$

Expected graphs

- **Resistor:** straight line through origin (Ohm's law).
- **Filament lamp:** curve — resistance increases with temperature.
- **Diode** (not part of this core practical, but you must explain its I–V graph, 10.18): conducts above ≈ 0.6 V forward bias only; nearly no current otherwise.

Part b — series and parallel circuits

1. Build a series circuit of two filament lamps (or resistors) with an ammeter. Measure the current at different points and use a voltmeter to measure the p.d. across each lamp and across the supply.
 2. Build a parallel circuit with the same components, with ammeters in each branch and in the main circuit, and voltmeters across each lamp.
 3. Compare the readings, and use $V = I \times R$ to work out the resistance of each lamp and of the whole circuit.
- **Series:** current is the same everywhere; the supply p.d. is shared between the components; total resistance is the sum of the individual resistances.
 - **Parallel:** p.d. across each branch equals the supply p.d.; the branch currents add up to the total current; total resistance is less than the smallest branch resistance.

⚠ Safety: Keep currents low and switch off between readings — resistors and lamps become hot.

CP6 (14.3) — Investigating the densities of solids and liquids

Apparatus: Balance, vernier callipers (or ruler), measuring cylinder, eureka can, beaker, water, selection of regular and irregular solids.

Method (regular solid)

1. Measure mass on a balance.
2. Measure dimensions (length $\times$ width $\times$ height) to find volume.

3. $\rho = m / V$.

Method (irregular solid)

1. Measure mass.
2. Fill eureka can to spout with water; immerse the object fully; collect the displaced water in a measuring cylinder.
3. Volume displaced = volume of object; calculate ρ .

Method (liquid)

1. Measure mass of empty measuring cylinder.
2. Add a known volume of liquid; measure new mass.
3. Mass of liquid $\div$ volume = density.

$$\rho = m / V$$

Exam Tip: Take readings from the bottom of the water meniscus at eye level. Measure dimensions 3× and mean to reduce random error.

CP7 (14.11) — Properties of water: specific heat capacity of water and a temperature–time graph for melting ice

Apparatus: Beaker (insulated with a lid), water, balance, low-voltage immersion heater and power supply, joulemeter (or ammeter, voltmeter and stopwatch), thermometer; crushed ice, boiling tube or small beaker, larger beaker of water as a water bath, Bunsen burner, tripod and gauze, stopwatch.

Method — specific heat capacity of water

1. Measure the mass m of the water: weigh the empty beaker, add water and weigh again.
2. Insulate the beaker and fit a lid. Put the immersion heater and thermometer into the water.

3. Record the initial temperature T_1 and zero the joulemeter.
4. Switch on the heater. Stir gently and heat until the temperature has risen by about 10–20 °C.
5. Switch off; the temperature may continue to rise briefly — record maximum T_2 and the joulemeter reading E .
6. Specific heat capacity $c = E / (m \times \Delta T)$, where $\Delta T = T_2 - T_1$.

$$\Delta Q = m \times c \times \Delta \theta$$

Method — melting ice

1. Put crushed ice in a boiling tube with a thermometer and record its starting temperature (below 0 °C if taken straight from a freezer).
2. Stand the tube in a beaker of water and heat the water bath gently with the Bunsen burner.
3. Stir and record the temperature every 30 s until all the ice has melted and the water has warmed for a few more minutes.
4. Plot temperature against time.

Analysis

- The accepted value for water is about 4200 J/kg °C.
- The melting graph rises to 0 °C, stays flat while the ice melts, then rises again. On the flat section energy is still being supplied, but it breaks bonds between particles (latent heat of fusion) instead of raising the temperature.

Errors and Improvements

- Some energy is lost to the surroundings → calculated c is **too high**.
- Improve with insulation, a lid, and a smaller ΔT rise.
- Stir the water so the thermometer reads the average temperature, not a hot spot near the heater.

⚠ Safety: Heater gets hot — don't touch it when live, and only switch it on when it is fully under water. Take care with the Bunsen burner and hot water bath; wear eye protection.

CP8 (15.6) — Investigating extension and work done when applying forces to a spring

Apparatus: Clamp stand, helical spring, metre rule, pointer, 100 g slotted masses (and hanger), safety mat, set square.

Method

1. Hang the spring vertically from a clamp; attach a pointer at the bottom.
2. Read the initial position of the pointer against the metre rule.
3. Add a mass (known weight = mg), wait for it to settle, and read the new pointer position.
4. Calculate extension (x) = new position – initial position.
5. Repeat, adding masses in equal increments. Record at least 8 pairs (force, extension).
6. Plot force (y -axis) against extension (x -axis).

$$F = k \times x \quad E = \frac{1}{2} \times k \times x^2$$

Analysis

- Straight line through origin → obeys Hooke's law. Gradient = spring constant k .
- Line curves beyond the **limit of proportionality**.
- Work done in stretching the spring (energy stored) = area under the force–extension graph. In the straight-line region this equals $\frac{1}{2} \times k \times x^2$ (extension in metres).

⚠ Safety: Use a safety mat under the masses; wear eye protection; clamp the stand to the bench so it does not topple.

Exam Tip: Use a set square to read the pointer horizontally — avoids parallax. Use 0.10 N (10 g) masses for small springs for more data points.

Not a core practical for 1PH0 — Investigating thermal insulators

This is not one of Edexcel's eight core practicals, but it is useful practice for specification points 3.9–3.10 (reducing unwanted energy transfer by thermal insulation; how wall thickness and thermal conductivity affect the rate of cooling).

Apparatus: Metal can / copper calorimeter, different insulating materials (bubble wrap, wool, cotton, foam), hot water, thermometer, stopwatch, lid.

Method

1. Pour a fixed volume of hot water (e.g. 200 ml at 80 °C) into the calorimeter.
2. Wrap it in one of the insulating materials.
3. Record the temperature every minute for 10 minutes with a lid on.
4. Plot temperature against time; compare the gradient for each material.
5. The material with the smallest rate of temperature decrease is the best insulator.

Control Variables

- Volume and starting temperature of water;
- Thickness of insulation layer;
- Shape and material of the calorimeter;
- Surrounding room temperature.

Exam Tip: For a valid comparison, change only the **insulating material**. Any difference in thickness, water volume or can shape will invalidate the test.

Key Practical Skills

- Identify dependent, independent and control variables.
- Plot graphs with labelled axes, units, and sensible scales; draw lines of best fit.

- Estimate uncertainties: resolution of ruler / thermometer / stopwatch.
- Explain random vs systematic errors and how you reduce them.
- Improve methods: insulate, use light gates / data loggers, take more repeats.
- Name the equipment precisely — e.g. "eureka can", "vernier callipers", "joulemeter".